

Quoc Chuong NGUYEN

CURRICULUM VITAE

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RESEARCH INTEREST

Mathematics : Game Theory · Network Science · Category Theory
Computer Science : Complexity Theory · Reinforcement Learning
Quantum Computation : Quantum Algorithm · Quantum Artificial Intelligence

EDUCATION & TRAINING

Education

- 2023 - 2025 **Master of Arts in Mathematics at SUNY, University at Buffalo**
(State University of New York at Buffalo), New York City, the US
Track: Applied Mathematics
CGPA: 3.889/4.0 and **Degree GPA: 3.967/4.0** (Grade A/A-: High Distinction)
Conferral Date: June 1, 2025
- 2014 - 2018 **Bachelor of Science in Physics at University of Science (HCMUS)**
Vietnam National University, Ho Chi Minh City (VNU-HCM), Vietnam
Track: Theoretical Physics
CGPA: 8.27/10 or 3.5/4.0 (top 2% over 300 students)
Graduated Seminar: Simulation of Quantum Many-body Dynamics
using the Monte-Carlo algorithm with Python (Non-Thesis option)

PUBLICATION

Published

- **(Co-author) Chuong et al.** “FQsun: A Configurable Wave Function-Based Quantum Emulator for Power-Efficient Quantum Simulations”, IEEE Access, vol. 13, pp. 93271-93286, 2025.
- **(First-author) Chuong et al.** “Qsun: an open-source platform towards practical quantum machine learning applications”, Machine Learning: Science and Technology, vol. 3, pp. 015034, 2022.

Preprints

- **(First-author) Chuong.** “Network Sampling: An Overview and Comparative Analysis”, arXiv:2504.17701, 2025.
- **(Co-author) Chuong et al.** “Hybridising Reinforcement Learning and Heuristics for Hierarchical Directed Arc Routing Problems”, arXiv:2501.00852, 2025.
- **(First-author) Chuong.** “Monte Carlo Analysis of Boid Simulations with Obstacles: A Physics-Based Perspective”, arXiv:2412.10420, 2024

- **(First-author) Chuong** and Trung Kien Le. “Toward a comprehensive simulation framework for hypergraphs: a Python-base approach”, arXiv:2401.03917, 2024.

ACADEMIC EXPERIENCE

Recent Remote Collaborations

- May 2024 - Now: In collaboration with Prof. Hy Truong Son (University of Alabama at Birmingham, US) and Prof. Thu Huong Dang (Lancaster University, England)
 - Combinatorial Optimization, Operation Research: Solving the Hierarchical Directed Capacitated Arc Routing Problem with heuristic algorithms

Software

- QSun: Quantum platform to simulate quantum circuits using NumPy array
 - https://github.com/ChuongQuoc1413017/Quantum_Virtual_Machine
- Quantum Evolutionary Algorithm: Optimize quantum circuits' structure using Evolutionary Algorithms
 - https://github.com/ChuongQuoc1413017/Quantum_Evolutionary_Algorithm
- HyperRD: Simulation and analysis of random and dynamical hypergraph
 - https://github.com/ChuongQuoc1413017/Hypergraph_RD

Lecture Notes

- Summary of Lie Algebra in Particle Physics (Vietnamese)
 - <https://github.com/ChuongQuoc1413017/Note/blob/main/Lie%20Algebra.pdf>
- Quantum Machine Learning at 〈QCS | 2021〉
 - https://github.com/ChuongQuoc1413017/Note/blob/main/QML_Lecture_Note.pdf

Research Positions

Institute of Fundamental & Applied Sciences, Duy Tan University July 2025 - Now

Website: <https://ifas.duytan.edu.vn/>

Position: Researcher

Job Description: develop quantum machine learning models and quantum algorithms

QuantumLab - Ho Chi Minh City Institute of Physics, Vietnam Sep 2021 - May 2022

Website: <https://lantrann.github.io/QuantumLab-HCMIP>

Position: Affiliated Member

Job Description: develop quantum machine learning applications in chemistry and quantum metrology

Research Assistant for Dr. Hung Q. Nguyen, Vietnam Nov 2019 - Apr 2023

Position: Research Assistant

Job Description: develop Qsun (Quantum Computing simulator, Python) and Variational Quantum Eigensolver's applications in machine learning

Teaching

Teaching Assistant (TA) at HCMUS, Vietnam Description: Instructor in Physical-Informed Neural Networks class	Sep 2025 - Dec 2025
Mentor of Directed Reading Project program at SUNY, UB Description: teach undergrad students about Computational Thinking and Data Structures & Algorithms	Jan 2024 - May 2024
Teaching Assistant (TA) at SUNY, UB Description: TA in Calculus, Linear Algebra, and Stochastic Processes	Aug 2023 - May 2025

TALKS/POSTERS AT CONFERENCES, WORKSHOPS, AND SCHOOLS

"Introduction to Quantum Algorithms" at Quantum Winter School, HCMUT, Vietnam	22 Nov 2025 Lecturer
"Quantum Machine Learning & Qsun platform" at PIPC 2025: Celebrating 100 years of quantum physics, Vietnam	13 Oct - 15 Oct 2025 Poster
"Quantum Machine Learning" at Quantum Computing Seminar, Vietnam	19 Aug 2025 Speaker
"Quantum Machine Learning" at Quantum Computing School - 〈QCS 2021〉, Vietnam	04 Dec - 12 Dec 2021 Contributory Lecturer
"Half-wormhole solutions & Black Hole singularity" at The 46th Vietnam Conference on Theoretical Physics (VCTP-46)	13 Oct 2021 Poster

HONORS & AWARDS

Tuition & Teaching Assistantship (SUNY, University at Buffalo) Description: for Master/Ph.D. student with a good profile	Aug 2023 - June 2025 2 academic years
Academic incentive scholarships (University of Science, HCMUS) Description: for students with the highest GPA and contributions to extracurricular activities for society each semester.	Nov 2014 - Nov 2018 4 academic years

CONFERENCES, WORKSHOPS, AND SCHOOLS ATTENDED

VIASM-ICTP Summer School in Group Theory and Representation Theory	9 - 21 Aug 2022
The Mathematics of Interactive Bose Gas 2022	1 - 6 Aug 2022
Topological Quantum Electrons Interacting In-persons	10 - 16 July 2022
Spring School on Superstring Theory and Related Topics	9 - 13 May 2022
The 10th International Workshop on Solid-State Quantum Computing	29 Nov - 1 Dec 2021
The 2021 Vietnam Operations Research Network Meeting	26 - 27 Nov 2021
Vietnam School on Neutrinos - VSON2021	29 Aug - 09 Sep 2021
Vietnam - USA Joint Mathematical Meeting - VNUS 2019	10 - 13 Jun 2019
Asia-Europe-Pacific School of High-Energy Physics, Vietnam	12 - 25 Sep 2018

REFeree FOR PEER-REVIEWED JOURNALS

- [Quantum Machine Intelligence](#), [Scientific Reports](#), [SoftwareX](#)

WORKING EXPERIENCE

Education

FUNiX Co. Ltd., Vietnam

Jul 2020 - Nov 2021

Position: Collaborative Assistant

Job Description: develop machine learning & data science courses, translate courses' contents from Vietnamese to English, fix coding bugs

Saigon Scientists Co., Vietnam

Jul 2018 - Jul 2019

Position: Research & Development Assistant

Job Description: develop STEM lessons, translate courses' contents from Vietnamese to English, project management

LANGUAGES & COMPUTER SKILLS

VIETNAMESE: Mother tongue

ENGLISH: Intermediate

PROGRAMMING LANGUAGES: Proficient: Python, NetworkX, Qiskit, PennyLane
Basic: Mathematica, Matlab, R, LaTeX, Haskell, Lean

CERTIFICATIONS

- GRE General Test: 317/340 (75th percentile)
 - Quant: 167/170 (87th percentile), Verbal: 150/170, AW: 3.5/6.0
- [Introduction to Discrete Mathematics for Computer Science Specialization \(offered by University of California San Diego\)](#) on *Mathematical Thinking, Probability, Combinatorics, and Graph Theory* at [Coursera](#) in 2022, Credential ID: ZDL-RFFSJ6H8N.
- [Introduction to Logic and Critical Thinking Specialization \(offered by Duke University\)](#) on *Argument Analysis, Deductive Reasoning, Inductive Reasoning, and Fallacies* at [Coursera](#) in 2022, Credential ID: PNT6RQ3G9Q89.
- [Mind and Machine Specialization \(offered by University of Colorado Boulder\)](#) on *Cognitive Science and Artificial Intelligence* at [Coursera](#) in 2022, Credential ID: HNTZFNDS4LU7.
- [Writing in the Sciences \(offered by Stanford University\)](#) at [Coursera](#) in 2022, Graded 100/100 with Honors, Credential ID: HTJ5EJQ2VZ2F.
- [Game Theory \(offered by Stanford University and The University of British Columbia\)](#) at [Coursera](#) in 2022, Graded 100/100, Credential ID: SUFX9JZ2YTSA.
- [Research Writing in the Sciences](#) course at [INASP, the United Kingdom](#) (International Network for Advancing Science and Policy) in 2022, Graded Merit.
- [Machine Learning](#) courses on *Regression, Classification, and Clustering* at [FUNiX, Vietnam](#) (an online learning platform for Vietnamese) in 2020, Graded 10/10.
- [ERASMUS+](#) course on *Symmetry and Invariance in Physics* (a short course about symmetries and Noether's theorem taught by [Prof. Amaury Mouchet](#) from François Rabelais University, France, and organized at HCMUS) in 2016, Non-Graded.

REFERENCES

Prof. Thu Huong Dang ([Google Scholar](#))
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